Quality-Adaptive Loss Reweighting Using Expert-Assigned Image Quality for Robust Skin Lesion Classification

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***Abstract*—Diagnostic image quality varies widely in clinical practice, yet quality-aware training methods have had to estimate quality from image statistics or feature norms rather than observe it, because expert judgments of image quality are almost never recorded alongside diagnostic labels. This paper has proposed quality-adaptive loss reweighting, which scales each training sample's classification loss by a real expert-assigned rating of its diagnostic image. Two opposing directions have been evaluated: down-weighting low-quality samples as less reliable, and up-weighting them to force the model to learn from difficult images. The second, the opposite of what the closest face recognition precedents apply, has improved balanced accuracy by 0.037 and malignant sensitivity by 0.092 over an identical baseline. Because a gain of this size is comparable to the fold-to-fold standard deviation, we have validated it causally rather than statistically: a shuffled-quality control, preserving the weight distribution but permuting the lesion-to-rating assignment, has removed the gain entirely and returned performance to baseline, establishing that the improvement has depended on the quality signal's information content rather than on generic reweighting. The full pipeline has reached 0.840 balanced accuracy, of which 65% of the total gain has been attributable to the quality-adaptive loss and the remainder to generic evaluation-time averaging. The method has been made possible by MCR-SL, the only public skin lesion dataset recording per-image expert quality ratings, on which we have also reported the first published benchmark and six robustness analyses, three of which have reversed an apparent positive result.**

***Index Terms*—skin lesion classification, multimodal fusion, dermoscopy, image quality, metadata fusion, deep learning, medical imaging benchmark**

I. Introduction

Medical images acquired in routine practice vary substantially in diagnostic quality. Lighting, focus, framing and hair occlusion all degrade the evidence a classifier receives, and training as though every image were equally informative assumes a uniformity clinical data does not have. Face recognition has addressed this by making the objective quality aware: AdaFace [14] and MagFace [15] adapt each sample's margin from a feature-norm estimate of image quality, both reducing emphasis on difficult low-quality samples. That reliance on an estimated proxy is a necessity, not a preference, since face datasets do not record what an expert thought of each photograph.

Skin lesion classification has followed a parallel path: convolutional networks reached dermatologist-level accuracy on large single-modality archives [2], [3], and a smaller line of work fused images with patient metadata [4], [5]. In neither case was a per-image expert judgment of quality available, so quality remained something to infer rather than observe.

MCR-SL [1] removes that constraint, and is the reason the proposed method is possible at all. It pairs 779 clinical and 1352 dermoscopic images across 240 lesions from 60 subjects with twenty-two subject-level metadata fields and, distinctively, with a one-to-ten quality rating that dermatologists assigned to the specific image used for each lesion's diagnosis. It is the only public skin lesion dataset carrying such a field, which makes it the enabling dataset for a loss driven by real expert quality rather than an estimated proxy, not merely a convenient testbed.

This paper therefore asks two separable questions: whether performance degrades on lower-quality images, and whether the objective can use the quality signal to improve robustness. The answers differ, the first a null and the second a positive validated by a causal control.

A. Research Gap

No prior work has used a direct expert rating of image quality as a loss-reweighting signal in medical image classification. Quality-aware objectives have relied on estimated proxies, quality-aware medical methods have acted on the architecture rather than the loss, and the closest loss-weighting scheme on skin lesions derives its difficulty signal from the model rather than from a human judgment of the photograph. The literature review that follows substantiates each of these in turn.

B. Contributions

This paper has made the following contributions.

- **Quality-adaptive loss reweighting:** a scheme that scales each sample's classification loss by a real expert-assigned image-quality rating. The direction that has worked, up-weighting low-quality samples, is the opposite of the face recognition precedents that motivated it, and we have offered a domain-specific explanation for why.

- **Causal validation via a shuffled-quality control:** permuting the lesion-to-rating assignment while preserving the weight distribution has removed the entire gain, establishing that it has depended on the quality signal's information content and not on generic per-sample reweighting.
- **A first published benchmark on MCR-SL:** evaluated under a subject-disjoint stratified five-fold protocol, together with a sweep of eleven follow-up configurations that has isolated which interventions help at this scale and which do not.
- **Six robustness analyses, including three reversed findings:** enabled by MCR-SL's quality ratings and partial histopathology confirmation, and reported together with the three cases in which an apparent positive finding has not survived a control.

II. Literature Review

Image-only dermoscopic classification has been studied extensively since Esteva *et al.* [2] matched dermatologist performance on a curated archive, with HAM10000 [3] becoming a standard benchmark. These studies have treated the image as the sole input.

Metadata fusion is a distinct thread. Pacheco and Krohling [4] have proposed the Metadata Processing Block, which reweights image features using patient metadata, evaluated on PAD-UFES-20 [5] (2298 clinical images, 1641 lesions). That dataset is the closest comparator in problem shape and is our primary external reference point, discussed later.

Quality-aware training has been developed most fully in face recognition. AdaFace [14] adapts the angular margin using the feature norm as a quality proxy, reducing emphasis on hard examples when an image is estimated to be low quality, since a degraded and difficult face is likely unidentifiable and forcing separation would fit noise. MagFace [15] ties feature magnitude to quality similarly. Both rely on an estimated proxy rather than a recorded human judgment.

Within medical imaging, quality has more often been handled architecturally. QGDR [17] routes fundus images through quality-conditioned expert branches for diabetic retinopathy grading, modifying the architecture rather than the loss, and derives its quality states largely from predicted pseudo-labels rather than from expert annotation.

Closest to this work, Hu and Yang [16] apply knowledge-based loss weighting on HAM10000, weighting at the class level for minority classes and at the instance level for challenging samples. Their instance-level term shares this paper's direction, but derives difficulty from the model and data themselves; the signal used here is external to both, an expert's judgment of the photograph recorded independently of any model. Our fusion module draws on Squeeze-and-Excitation [6] and is an established component, not a contribution.

III. Proposed Methodology

Fig. 1 summarizes the architecture. This paper has proposed a channel-gated fusion method for cross-modal skin lesion classification and has compared it against two baselines under an identical training and evaluation protocol.

A. Image Encoder

The upper stream of Fig. 1 encodes the dermoscopic image. An ImageNet-pretrained EfficientNet-B0 [7], [8] is trained end to end with no frozen layers, and a single forward pass returns both the final convolutional feature map, of size $1280 \times 7 \times 7$, and its globally average-pooled 1280-dimensional summary. The feature map is what the fusion stage gates; the pooled vector is used only by the concatenation baseline.

B. Metadata Encoder

The lower stream encodes patient and lesion context. Seventeen categorical fields each receive their own embedding of twelve dimensions, sized to the field's cardinality plus one reserved index for missing values, so missingness is represented explicitly and never imputed. Four numeric fields are z-scored using training-fold statistics only, and each is paired with a binary indicator recording whether it was present. Concatenating these gives a 212-dimensional vector, which a two-layer perceptron maps to the 128-dimensional metadata representation.

C. Channel-Gated Fusion

Two fusion variants share a common 256-dimensional projection. The late-fusion baseline simply concatenates the pooled image vector with the metadata vector. The proposed channel-gated variant, drawn as $\otimes$ in Fig. 1, instead maps the metadata vector through a linear layer and a sigmoid to a 1280-dimensional gate in $[0, 1]$, multiplies the convolutional feature map channel by channel by that gate, and only then pools. Patient context therefore modulates which visual channels survive pooling rather than being appended after it. This is the Squeeze-and-Excitation mechanism [6] conditioned on metadata instead of on the feature map's own statistics, and it is used here as an established component rather than claimed as a contribution.

D. Prediction Heads

Two heads operate on the fused 256-dimensional representation. The binary head is a single linear layer followed by a sigmoid, giving $P(\text{malignant})$, and is the only head whose output is reported as a headline result. A nine-class auxiliary head predicts the unified diagnosis under a class-weighted cross-entropy contributing at $0.4\times$, and is reported as an exploratory table only, since several of its classes contain fewer than ten lesions.

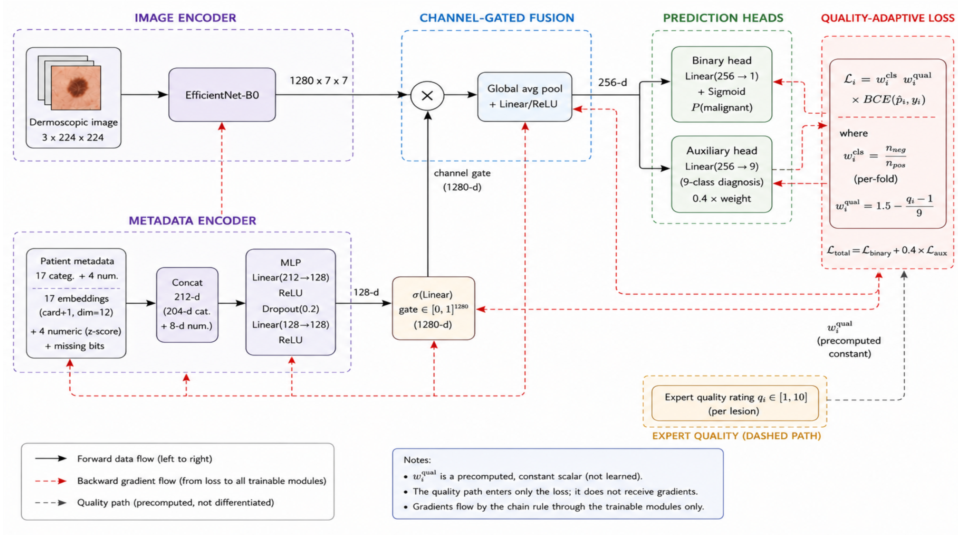

Fig. 1. Overview of the proposed architecture and training objective. The image and metadata streams are encoded separately and meet at a metadata-conditioned channel gate, which rescales the convolutional feature map channel by channel before pooling and projection. Solid arrows denote the forward path and the red path gradient flow; the expert quality rating enters only through the dashed path, where w_i^{qual} is a precomputed constant and is never differentiated.

E. Quality-Adaptive Loss

The contribution modifies neither the architecture nor the class of loss function, but the weight each sample carries within the existing objective. Let $q_i \in [1, 10]$ be the mean expert quality rating of lesion i's diagnostic image. A per-sample weight w_i^{qual} is applied alongside the per-fold class weight w_i^{cls}:

$$\mathcal{L}_i = w_i^{\text{cls}} \cdot w_i^{\text{qual}} \cdot \text{BCE}(\hat{y}_i, y_i). \tag{1}$$

Two opposing directions have been evaluated, each mapping the rating range onto $[0.5, 1.5]$:

$$\textit{trust}: \quad w_i^{\text{qual}} = 0.5 + (q_i - 1)/9, \tag{2}$$

$$\textit{hard\_mining}: \quad w_i^{\text{qual}} = 1.5 - (q_i - 1)/9. \tag{3}$$

The *trust* direction treats low-quality images as less reliable and down-weights them; *hard_mining* does the opposite, forcing the model to work harder on degraded images. Lesions without a valid rating receive a neutral weight of 1.0. The range is fixed rather than searched, and only these two directions are evaluated, so the comparison isolates direction rather than magnitude. Nothing unusual happens to the gradient: w_i^{qual} is a precomputed, non-differentiable scalar coefficient, not a learned parameter, so gradients propagate through the shared architecture by the standard chain rule from the weighted binary term.

Only *hard_mining* survives validation, as shown below, and its direction is the opposite of AdaFace's and MagFace's [14], [15]. One possible reason is that the settings degrade differently: a sufficiently degraded face can become unrecoverable, so emphasising it risks fitting noise, whereas in dermoscopy the diagnostic label was assigned from that same image, so the label remains valid and the sample remains an informative hard example.

This mechanism is distinct from the auxiliary quality-regression head noted earlier, which instead asks the model to *predict* the rating as a secondary output, adding a competing objective to a shared representation. Reweighting adds no parameters and changes only which samples dominate the gradient. The two are mechanistically different, and the auxiliary head's negative result stands separately rather than being superseded.

F. Training and Evaluation Protocol

Every configuration was evaluated under subject-disjoint stratified five-fold cross-validation, assigning each subject's lesions to exactly one fold so that no subject or near-duplicate image appeared in both a training and an evaluation fold, with folds balanced greedily on malignant-lesion count. Per rotation, one fold was the reported test fold and a second, disjoint fold served only for checkpoint selection by validation balanced accuracy; the test fold never influenced model selection, and no hyperparameter was tuned against final results.

Seven follow-up interventions were additionally evaluated under the identical protocol, the last two being evaluation-time only, and four stacked combinations bring the total to eleven configurations; these are described in the ablation study below.

IV. Experimental Results

A. Dataset

MCR-SL [1] comprises 240 lesions from 60 subjects, 42 malignant and 192 non-malignant, with 1352 dermoscopic and 779 clinical images, 28 lesions histopathology-confirmed, and 16 categorical plus 4 numeric metadata fields. Six lesions carry an undetermined malignancy label and are excluded, leaving 234 usable; The evaluation study below traces the further reduction to the 231 used in the analyses.

B. Evaluation Metrics

Six metrics are reported for every configuration: accuracy, sensitivity, specificity, macro-averaged F1, balanced accuracy, and the area under the receiver operating characteristic curve (AUROC). Each is aggregated across the five test folds as a mean and standard deviation.

Accuracy, the proportion of all lesions classified correctly, is a weak summary here: predicting every lesion non-malignant would already score above 80%.

Sensitivity, the proportion of malignant lesions correctly identified, is the clinically decisive measure and the one the proposed loss targets.

Specificity, the proportion of non-malignant lesions correctly cleared, constrains how far sensitivity can be bought, since each false alarm is an unnecessary biopsy.

Macro-averaged F1, the unweighted mean of the per-class harmonic means of precision and recall,

$$\text{macro-F1} = \frac{1}{2} \sum_{c \in \{0,1\}} \frac{2\,\mathrm{Pre}_c\,\mathrm{Rec}_c}{\mathrm{Pre}_c + \mathrm{Rec}_c}, \tag{4}$$

rewards doing well on both classes rather than the majority alone.

Balanced accuracy, the unweighted mean of the two class-wise rates,

$$\mathrm{BAcc} = \tfrac{1}{2}\left(\mathrm{Sen} + \mathrm{Spe}\right), \tag{5}$$

is used for checkpoint selection and as the primary comparison metric, correcting for the 17.9% malignant minority.

AUROC, the probability a randomly drawn malignant lesion is scored above a non-malignant one, is threshold-independent and so measures ranking quality separately from the decision boundary.

C. Existing vs. Proposed Methods

This is an internal comparison: every configuration in Table I was trained and evaluated by us under the identical protocol, on the same folds and lesions, so differences are attributable to the method alone. Channel-gated fusion reaches the highest macro-F1 but does not dominate: image-only has higher sensitivity (0.697 vs. 0.672) and AUROC (0.895 vs. 0.881), and balanced accuracy is effectively tied (0.784 vs. 0.783, against a standard deviation of 0.07–0.09). We report this rather than framing it as a clean win.

The quality-aware variant, which adds an auxiliary head predicting the rating, underperforms on five of six metrics (balanced accuracy 0.740 vs. 0.783, sensitivity 0.578 vs. 0.672) against a small specificity gain, trading a large sensitivity loss for a small specificity one, the worse direction for screening. This is a negative result for *predicting* quality, mechanistically distinct from the reweighting scheme introduced earlier.

Both weighting directions introduced earlier have been trained under the identical protocol, so the loss weighting is the only difference between them and the baseline. The *hard_mining* direction has improved every core metric: balanced accuracy by 0.037 (0.783 to 0.820), sensitivity by 0.092 (0.672 to 0.764), AUROC by 0.025 (0.881 to 0.906) and accuracy to 0.845, against a small specificity cost (0.895 to 0.875). The sensitivity gain is the clinically weightier one. The *trust* direction is approximately flat on balanced accuracy (0.788) and is not treated as an improvement. Since 0.037 is smaller than the fold-to-fold standard deviation of roughly 0.08, the comparison is underpowered, and we report that plainly: across the five paired folds *hard_mining* wins three on balanced accuracy (Wilcoxon $p = 0.81$) and four on AUROC ($p = 0.31$), and on sensitivity wins two while tying three, never losing a fold ($p = 0.18$). None approaches significance at $N = 5$, so the claim rests on the control below rather than on this comparison.

TABLE I
RESULTS OF EXISTING AND PROPOSED METHODS (MEAN ± STD, FIVE FOLDS). BOLD MARKS THE BEST VALUE IN EACH ROW

Metric	Image-only	Late fusion	Channel-gated
Accuracy	0.813 ± 0.020	0.831 ± 0.052	**0.833 ± 0.070**
Balanced acc.	**0.784 ± 0.073**	0.768 ± 0.090	0.783 ± 0.085
Macro F1	0.733 ± 0.038	0.743 ± 0.051	**0.757 ± 0.073**
Sensitivity	**0.697 ± 0.187**	0.644 ± 0.191	0.672 ± 0.159
Specificity	0.872 ± 0.061	0.891 ± 0.021	**0.895 ± 0.048**
AUROC	**0.895 ± 0.053**	0.851 ± 0.080	0.881 ± 0.065

Channel-gated, quality-aware variant: accuracy 0.812 ± 0.068, balanced accuracy 0.740 ± 0.065, macro F1 0.715 ± 0.065, sensitivity 0.578 ± 0.177, specificity 0.902 ± 0.073, AUROC 0.879 ± 0.041.

Causal validation via a shuffled-quality control. The obvious objection is that any per-sample reweighting might act as a regulariser, leaving the ratings incidental. We test this directly: within each fold's training portion the ratings are permuted across lesions, preserving the weight distribution exactly while destroying the lesion-to-rating correspondence. All else is identical, and evaluation always uses the true ratings.

Under this control, *hard_mining*'s balanced accuracy falls from 0.820 to 0.775 and its sensitivity from 0.764 to 0.667, both returning to the baseline's level (0.783 and 0.672).

A distribution-matched but uninformative weighting therefore recovers none of the gain, which establishes that the improvement depends on the weights tracking real quality rather than on reweighting as such. This is a causal argument rather than a statistical one, and it is why the result is reported despite an effect size comparable to the fold-to-fold standard deviation.

The control has also been informative negatively. The *trust* variant's one apparent advantage, a narrower accuracy gap between the highest and lowest quality terciles (0.082 against the baseline's 0.091), narrows further still under shuffling, to 0.021, while its balanced accuracy falls from 0.788 to 0.763: a weighting carrying no quality information produces a flatter profile than the mechanism intended to produce one. That advantage is therefore reported as a reversed finding rather than a second positive result.

D. Comparison with the Literature

Unlike the comparison above, this places the results against work by other authors on a different dataset. Nothing here was retrained by us and there is no shared test set, so Table II is context, not a controlled comparison.

TABLE II
EXTERNAL COMPARISON AGAINST THE CLOSEST PUBLISHED BINARY-TASK RESULT. NOT A CONTROLLED COMPARISON: DIFFERENT DATASET, DIFFERENT AUTHORS, NO SHARED TEST SET

Source	Task	N	Bal. acc.
This paper, channel-gated	binary	234 lesions	0.783
This paper, + *hard_mining*	binary	234 lesions	0.820
This paper, best configuration	binary	234 lesions	**0.840**
DiffMIC, PAD-UFES-20 [13]	binary	2298 images	0.836

PAD-UFES-20's six-class balanced accuracies (0.765–0.842) are not valid comparators for a binary task despite the shared metric name. The binary result we have verified directly is Uliana and Krohling's [13], 0.836 from a diffusion classifier over 2298 images under five-fold cross-validation, at the image level rather than our lesion level. This paper's best configuration has reached 0.840 ± 0.095. The nominal 0.004 difference is more than twenty times smaller than our own fold-to-fold standard deviation, so the two results are statistically indistinguishable and we have described ours as within noise of that benchmark rather than as exceeding it. The comparison is also not like for like: the two counts differ in unit, our 234 lesions against their 2298 images, and the underlying datasets differ by roughly sevenfold in lesion count and, more consequentially, in class balance, 17.9% malignant here against 47.5% there. HAM10000 [3] headline results have reached higher raw accuracy, but on a dermoscopy-only task with no metadata and roughly forty times more training images, and have not been treated as comparable.

MCR-SL has had no prior published benchmark, so this result is the current best on it by default; we have avoided framing that as a state-of-the-art claim, since there has been no prior number to surpass, and have emphasized the loss mechanism and its shuffled-quality validation as the principal contribution.

E. Ablation Study

Eleven configurations were evaluated under the identical protocol, each varying one component: the loss (focal loss [12], the *trust* direction), the input (hair removal and colour normalisation), the objective (a contrastive auxiliary term [11], the quality-regression head), the optimiser (decoupled weight decay with a cosine schedule [10], sharpness-aware minimisation [9]), and the evaluation (flip-averaged and multi-image test-time averaging), plus four stacked combinations. Sharpness-aware minimisation with test-time augmentation reached 0.822, the best of the sweep. Contrastive pretraining, the alternative optimiser and the quality-regression head each underperformed the baseline, and stacking every promising intervention together reached only 0.759.

Applying the two evaluation-time interventions on top of *hard_mining* rather than the plain baseline produces this paper's best configuration. Starting from the baseline's 0.783, the quality-adaptive loss reaches 0.820, test-time augmentation 0.833 with the highest sensitivity reported here at 0.808, and multi-image averaging 0.840 ± 0.095. It stacks cleanly here, whereas it reduced balanced accuracy alongside sharpness-aware minimisation in the sweep, so that non-additivity was specific to that combination.

The decomposition matters for how the headline is read. Of the 0.057 total gain, 0.037, roughly 65%, comes from the quality-adaptive loss and is causally validated by the control above; the remaining 0.020 comes from generic evaluation-time averaging unrelated to image quality. We state the split so the contribution is not overstated by the final figure.

F. Evaluation Study

These analyses use the channel-gated out-of-fold predictions over the 231 lesions carrying both a prediction and a quality rating; three of the 234 lacked a usable image in their assigned test fold and were excluded rather than imputed. Table III collects the numbers, and the items below give only the interpretation.

TABLE III
EVALUATION STUDY OVER THE 231 EVALUATED LESIONS. TERCILE SIZES ARE UNEVEN BECAUSE THE MEAN OF UP TO THREE INTEGER RATINGS TAKES FEW DISTINCT VALUES

Group	N	Baseline	*hard_mining*
Accuracy, low tercile	85	0.776	0.776
Accuracy, mid tercile	93	0.914	0.903
Accuracy, high tercile	53	0.868	0.868
Sensitivity, low tercile	23	0.667	**0.810**
Histopathology-confirmed	28	0.750	—
Panel-consensus only	203	0.867	—
Certainty vs. error	N	ρ	p
Pooled	231	−0.175	0.008
Non-malignant only	189	−0.127	0.083
Malignant only	42	−0.093	0.559
Consistent subjects (null 29.9)	55	27	0.954

Quality-stratified performance. The tercile pattern is not monotonic and the rating-error correlation is -0.093 ($p = 0.158$), so no strong quality-performance relationship is detected here, neither confirmed nor ruled out. *Hard_mining* does not flatten the profile; its benefit appears within terciles, as a large sensitivity gain on the lowest-quality one.

Reversed findings. Three apparently positive results did not survive scrutiny: *trust*'s narrower tercile gap, undercut by the shuffled control narrowing it further; the pooled certainty-error correlation, which dissolves once stratified by class; and a proposed change to train on all images per lesion, which on verification described what the pipeline already did. At this sample size an uncontrolled first look repeatedly produces positives that controls dissolve, and reporting that is part of the contribution.

Intra-subject consistency. Errors do not concentrate in particular patients: fewer subjects are perfectly consistent than the permutation null predicts, and none has all its lesions misclassified.

Ground-truth source. The model performs worse on histopathology-confirmed lesions, likely because histopathology is ordered more often for clinically ambiguous cases, making that subset harder by construction. With 28 lesions this is qualitative only.

Qualitative analysis. Gradients were taken with respect to the feature map *after* the metadata gate, with the metadata branch held fixed, so attribution reflects image evidence. Fig. 2 shows two lowest-tercile lesions the loss flips from a miss to a correct call: in the first, diffuse baseline attention contracts onto a single focus; in the second, attention moves from the image edge onto the lesion. The maps are 7×7 upsampled and indicate approximate attention only. These are illustrative rather than representative: across the 231 lesions the reweighting fixes six malignant lesions and breaks two, the net gain of four that a $+0.092$ sensitivity amounts to at this scale.

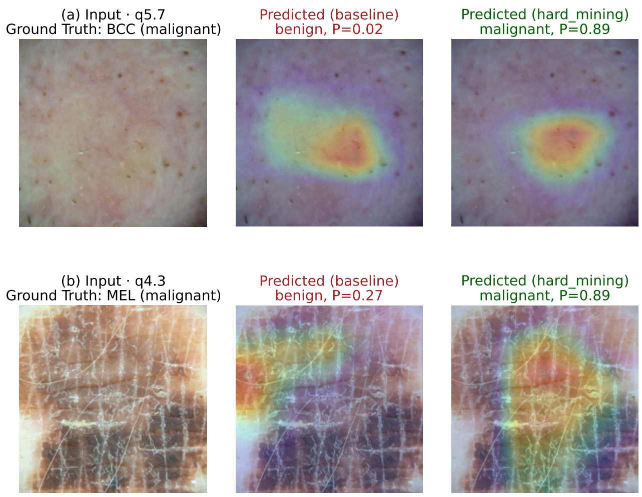

Fig. 2. Grad-CAM on the post-gate feature map for two lesions that quality-adaptive reweighting flips from a miss to a correct call.

V. Conclusion and Future Work

Real clinical images differ in how much diagnostic evidence they carry, and this paper asked whether an expert's judgement of that evidence can weight training. The proposed scheme multiplies each sample's classification loss by a coefficient derived from its expert quality rating. Of the two directions tested, up-weighting poor images proved effective, improving balanced accuracy by 0.037, sensitivity by 0.092 and AUROC by 0.025 over an identical baseline; evaluation-time averaging carried the pipeline to 0.840 ± 0.095, though only 65% of that gain belongs to the quality mechanism. That the useful direction inverts the face-recognition precedent [14], [15] is consistent with dermoscopy retaining a usable diagnosis under degradation where facial identity does not.

Whether this constitutes evidence deserves care. The improvement is not larger than the fold-to-fold spread and no paired fold-level test reaches significance; what supports the claim is the control, which returns the model to baseline once ratings are permuted. At 234 lesions we take causal evidence of this kind to be the appropriate standard. The two questions also separated: no reliable association was found between rated quality and error, nor between certainty and error once class composition was accounted for, yet the objective still extracted usable signal.

Constraints follow from the data: 234 usable lesions from one institution, 28 histopathology-confirmed, a single seed, and eight or nine positive cases per test fold. Next steps are to repeat the control across many permutations for an empirical p-value, and to test the mechanism on other datasets carrying expert quality annotation, DeepDRiD [18] being the most direct candidate.

References

[1] M. Castro-Fernandez *et al.*, "MCR-SL: A multimodal, context-rich skin lesion dataset for skin cancer diagnosis," *Data*, vol. 10, no. 10, art. 166, 2025.

[2] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, pp. 115–118, 2017.

[3] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, art. 180161, 2018.

[4] A. G. C. Pacheco and R. A. Krohling, "An attention-based mechanism to combine images and metadata in deep learning models applied to skin cancer classification," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 9, pp. 3554–3563, 2021.

[5] A. G. C. Pacheco *et al.*, "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," *Data in Brief*, vol. 32, art. 106221, 2020.

[6] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018.

[7] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.

[8] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.

[9] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, "Sharpness-aware minimization for efficiently improving generalization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[10] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.

[11] P. Khosla *et al.*, "Supervised contrastive learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 18661–18673.

[12] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[13] J. J. M. Uliana and R. A. Krohling, "Diffusion models applied to skin and oral cancer classification," 2025, arXiv:2504.00026. [Online]. Available: https://arxiv.org/abs/2504.00026

[14] M. Kim, A. K. Jain, and X. Liu, "AdaFace: Quality adaptive margin for face recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022.

[15] Q. Meng, S. Zhao, Z. Huang, and F. Zhou, "MagFace: A universal representation for face recognition and quality assessment," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021.

[16] M. Hu and X. Yang, "Attention-driven lightweight model for pigmented skin lesion detection," 2023, arXiv:2308.02119. [Online]. Available: https://arxiv.org/abs/2308.02119

[17] Y. Pan, X. Cai, P. Xiong, F. Mei, Z. Wang, and J. Hong, "From discarding to leveraging: quality-aware collaborative learning for robust diabetic retinopathy grading," *Frontiers in Medicine*, vol. 13, art. 1795484, 2026.

[18] R. Liu, X. Wang, Q. Wu *et al.*, "DeepDRiD: Diabetic retinopathy—grading and image quality estimation challenge," *Patterns*, vol. 3, no. 6, art. 100512, 2022.